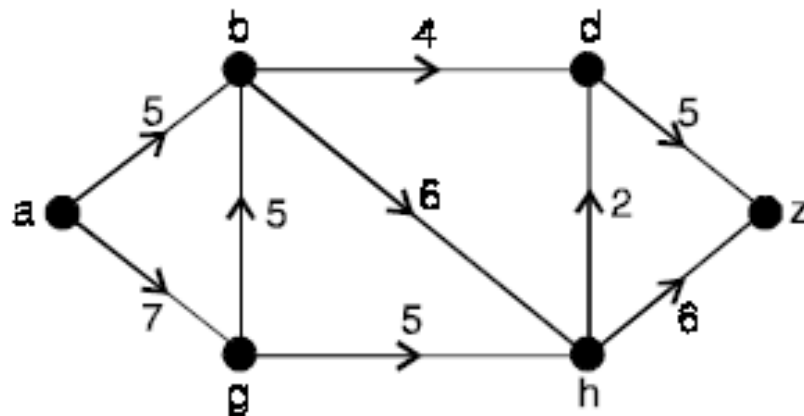


Transport Networks

A weighted directed graph is said to be a *transport network* if the following conditions are satisfied:

1. It is connected and contains no loops.
2. There is one and only one vertex in the graph that has no incoming edge.
3. There is one and only one vertex in the graph that has no outgoing edge.
4. The weight of each edge is a nonnegative real number.



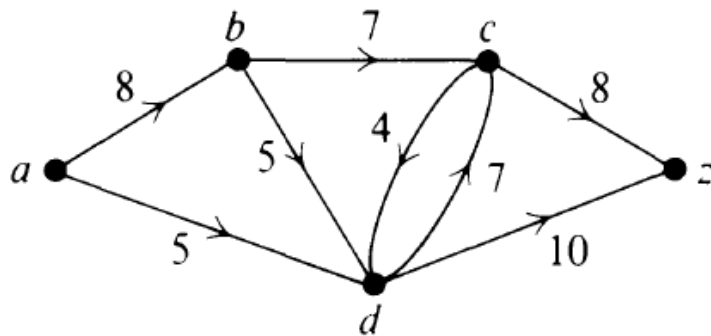
Transport Networks

- In a transport network, the vertex that has no incoming edge is called the source and is denoted by a , the vertex that has no outgoing edge is called the sink and is denoted by z .
- The weight of an edge is called the capacity of the edge. The capacity of the edge (i, j) is denoted by $w(i, j)$.

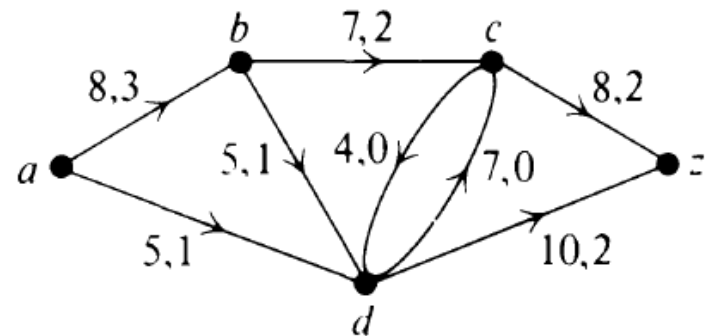
Flow in Transport Networks

A *flow* in a transport network, ϕ , is an assignment of a nonnegative number $\phi(i, j)$ to each edge (i, j) such that the following conditions are satisfied:

1. $\phi(i, j) \leq w(i, j)$ for each edge (i, j) .
2. $\sum_{\text{all } i} \phi(i, j) = \sum_{\text{all } k} \phi(j, k)$ for each vertex j except the source a and the sink z .†



(a)



(b)

Maximum Flow

- A maximum flow in the transport network is flow that achieves the largest possible value.

$$\phi_v = \sum_{\text{all } i} \phi(a, i) = \sum_{\text{all } k} \phi(k, z)$$

Above equation means that total outgoing flow at the source is equal to total incoming flow at the sink.

Cut in transport network

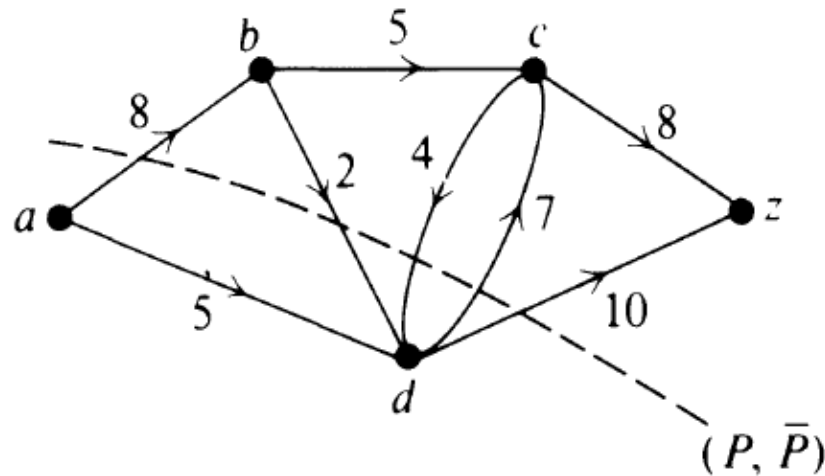
- A cut in the transport network is a cut set of the undirected graph, obtained from the transport network by ignoring the direction of the edges, that separates the source from the sink.

The notation $(P, \bar{P})$ is used to denote a cut that divides the vertices into two subsets P and $\bar{P}$, where the subset P contains the source and the subset $\bar{P}$ contains the sink. The *capacity of a cut*, denoted by $w(P, \bar{P})$, is defined to be the sum of the capacities of those edges incident from the vertices in P to the vertices in $\bar{P}$; that is,

$$w(P, \bar{P}) = \sum_{i \in P, j \in \bar{P}} w(i, j)$$

Cut in transport network

- $P = \{ a, d \}$, $P' = \{ b, c, z \}$
- Capacity of cut = $8+7+10 = 25$



Theorem

- The value of any flow in a given transport network is less than or equal to the capacity of any cut in the network.

Maximum Flow problem(Ford Fulkerson Algorithm)

- Ford-Fulkerson algorithm is a greedy approach for calculating the maximum possible flow in a network or a graph.
- We can visualize the understanding of the algorithm using a flow of liquid inside a network of pipes of different capacities. Each pipe has a certain capacity of liquid it can transfer at an instance. For this algorithm, we are going to find how much liquid can be flowed from the source to the sink at an instance using the network.

Terminologies Used

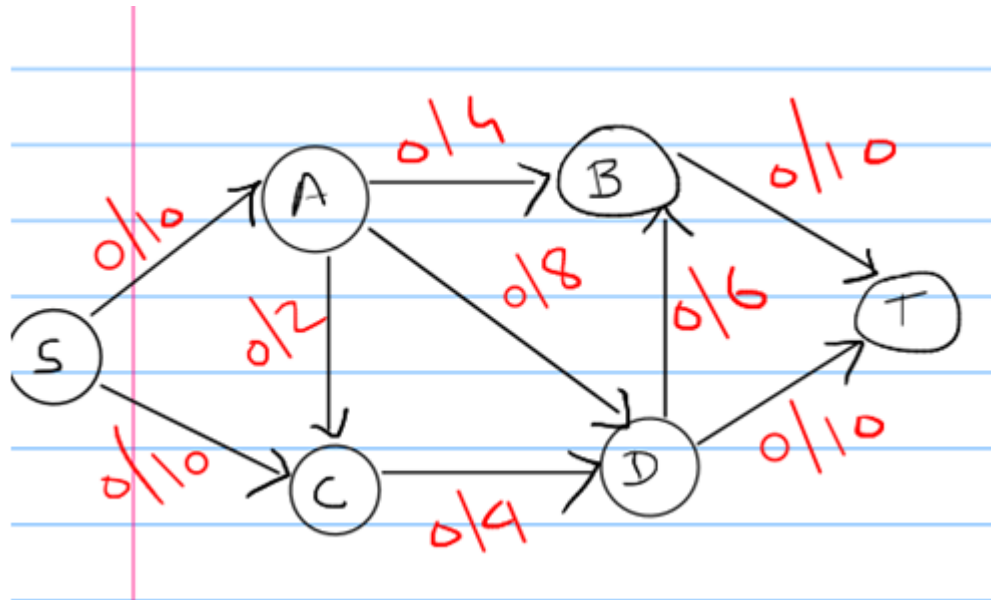
- **Augmenting Path**
 - It is the path available in a flow network.
- **Residual Graph**
 - It represents the flow network that has additional possible flow.
- **Residual Capacity**
 - It is the capacity of the edge after subtracting the flow from the maximum capacity.

Augmenting Path

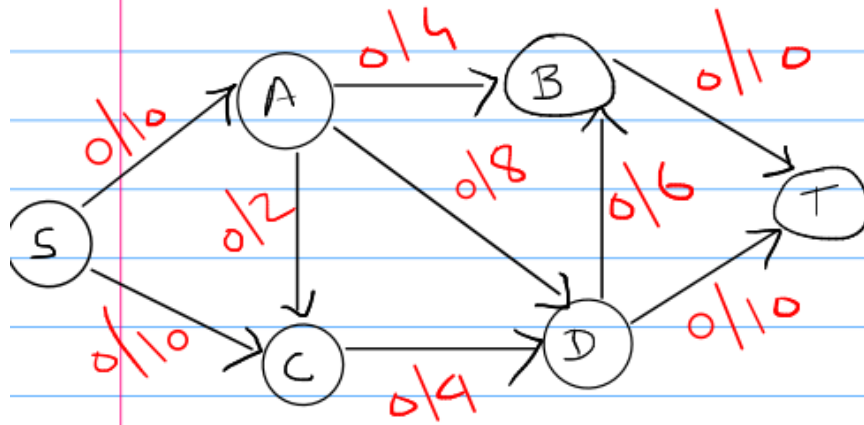
Augmenting paths in general

- increase flow on forward edge (if not full)
- decrease flow on backward edge (if not empty)

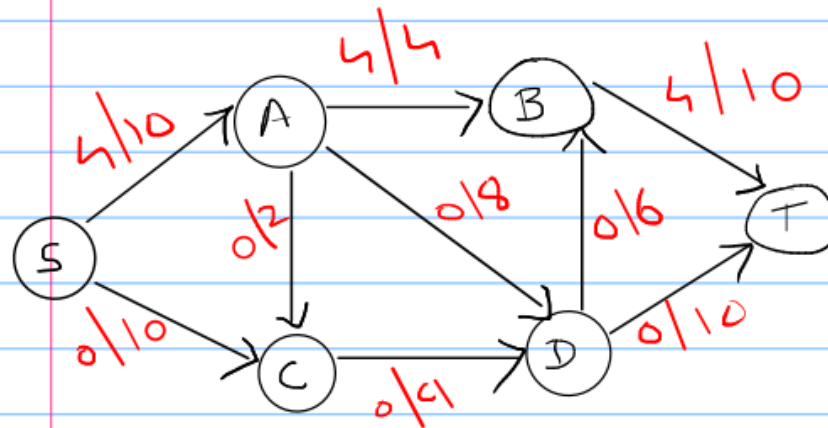
Example



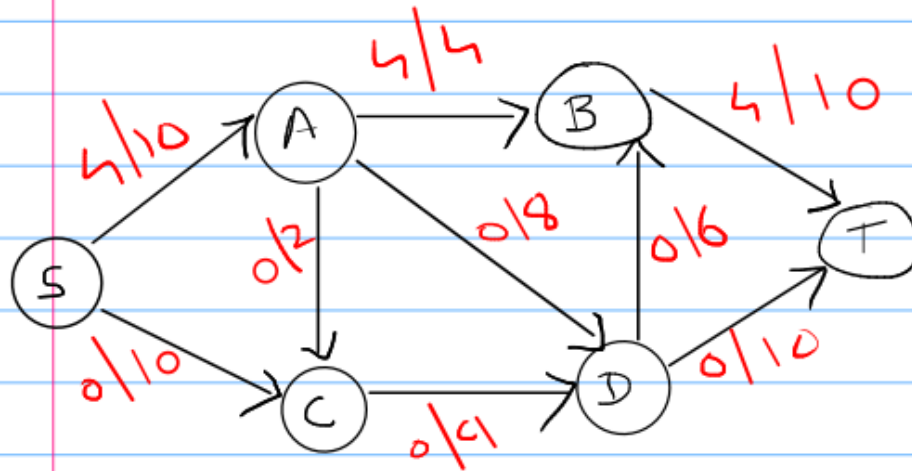
Step-1



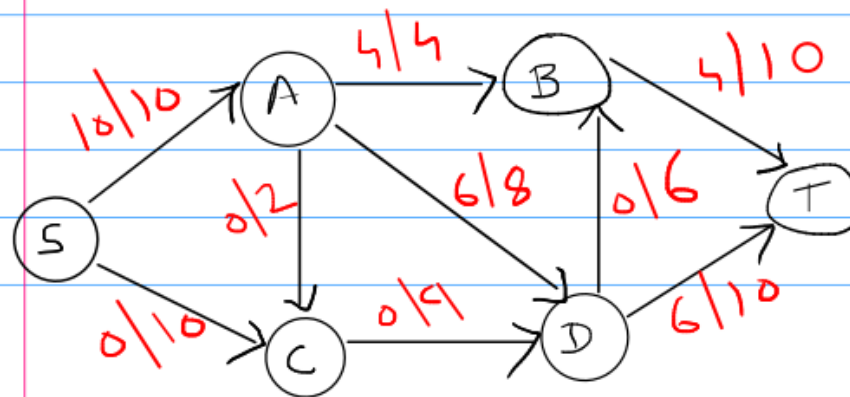
Augmenting path	Bottleneck Capacity
S-A-B-T	4



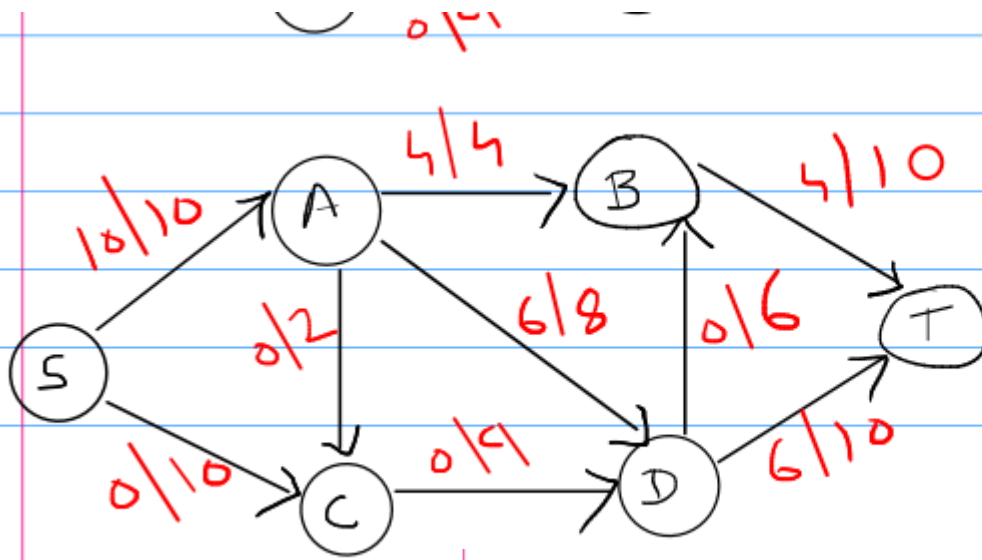
Step-2



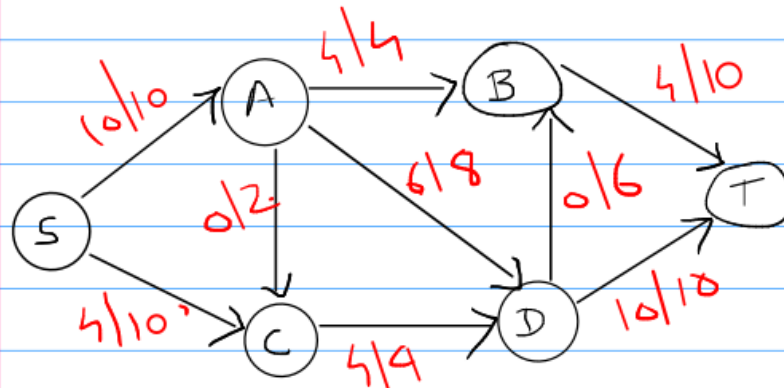
Augmenting Path	Bottleneck Capacity
S-A-B-T	4
S-A-D-T	6



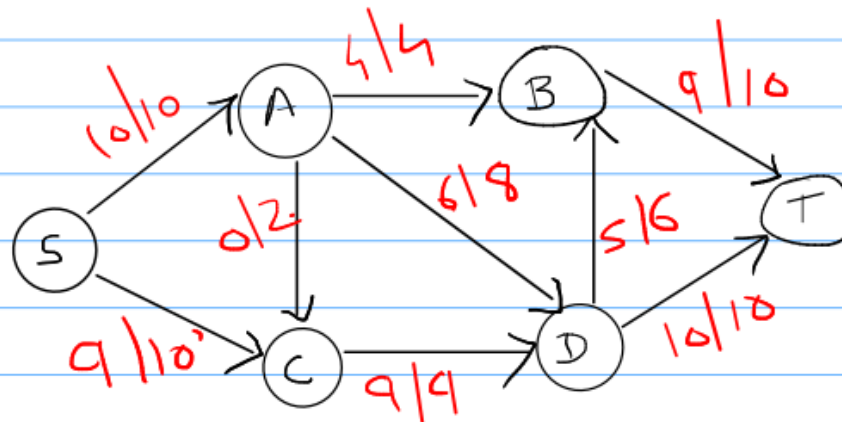
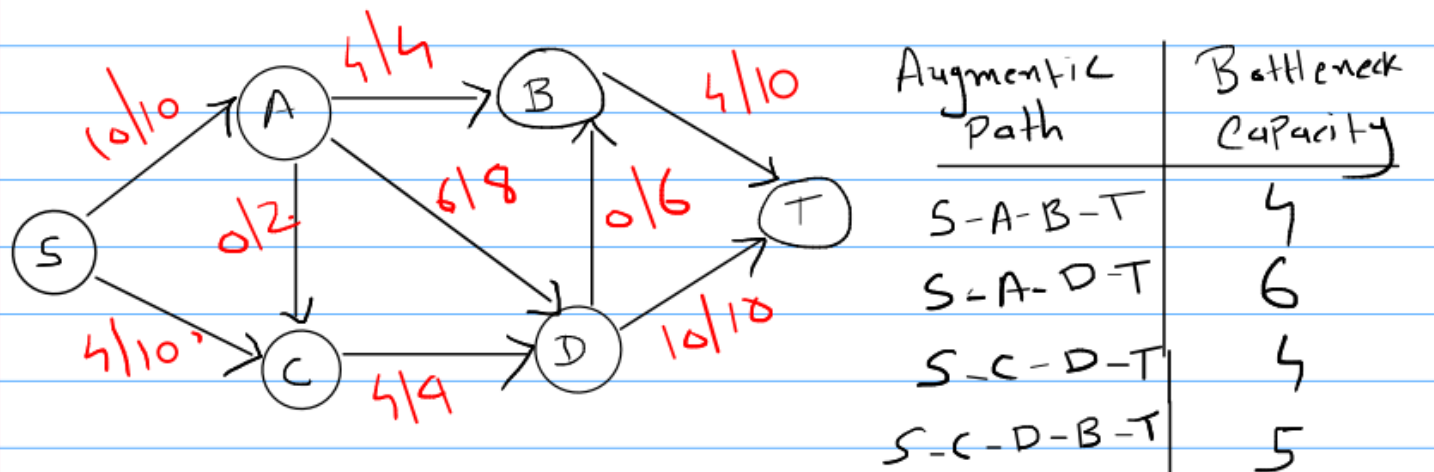
Step-3



Augmenting Path	Bottleneck Capacity
S-A-B-T	4
S-A-D-T	6
S-C-D-T	4



Step-4

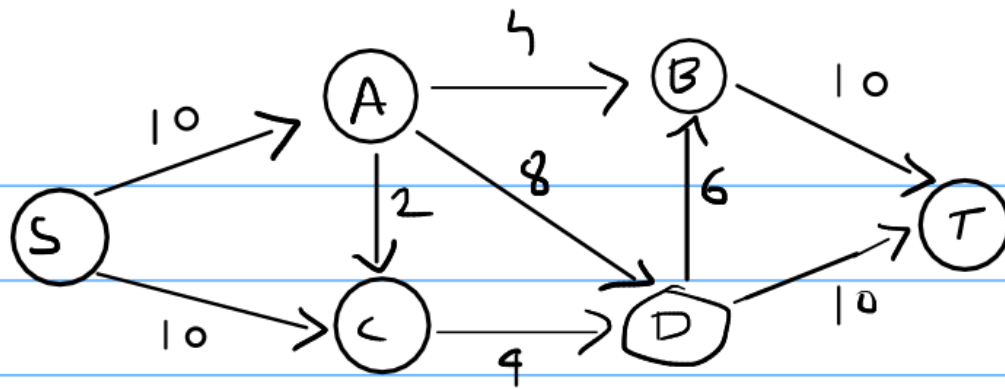


Step-5

- No further paths are possible, so the answer :

Augmenting Path	Bottleneck Capacity
S-A-B-T	4
S-A-D-T	6
S-C-D-T	4
S-C-D-B-T	5
Ans = 19	

Another solution

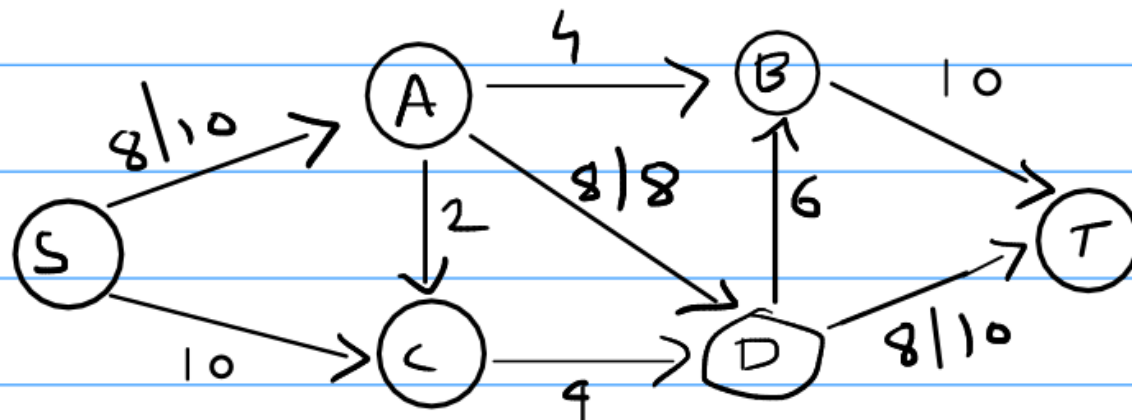


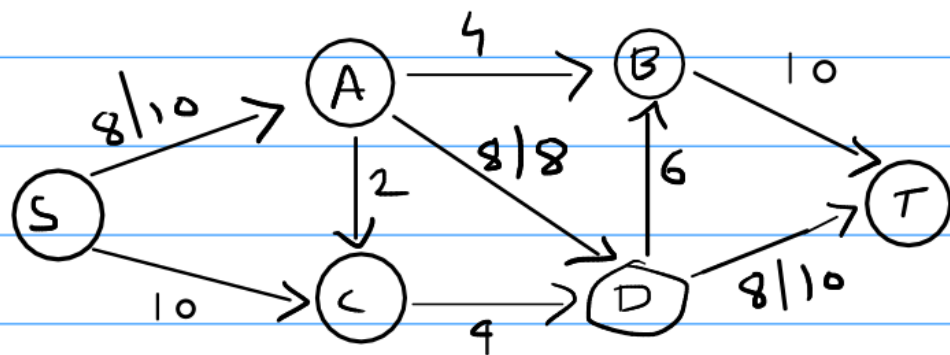
Augmenting
Path

S-A-D-T

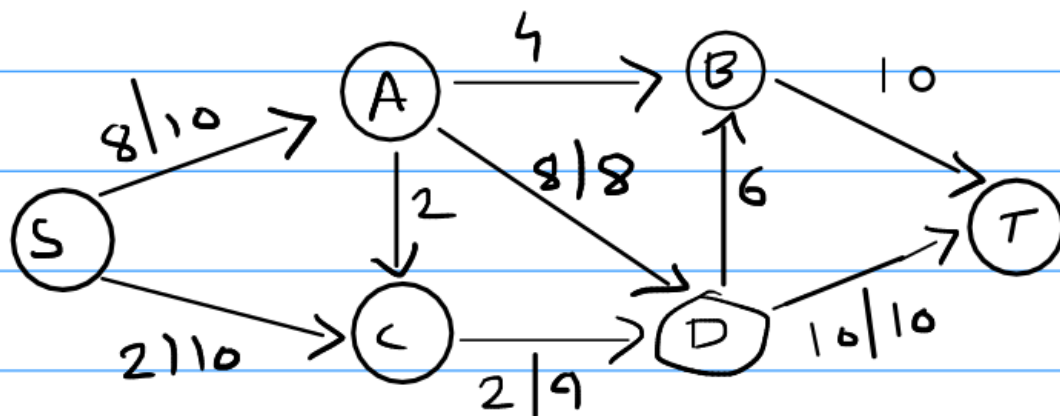
flow

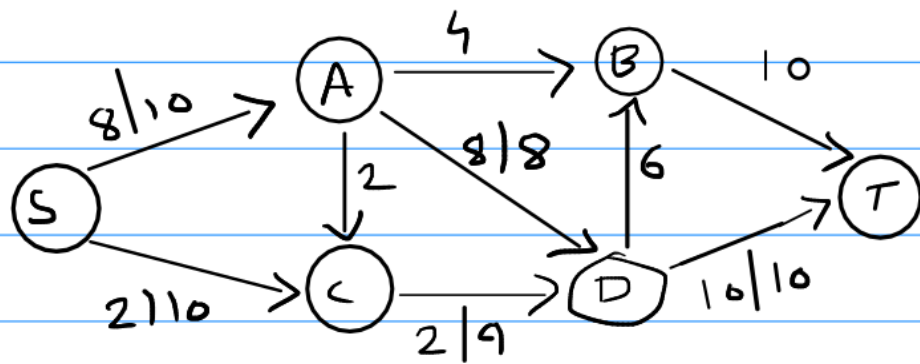
8



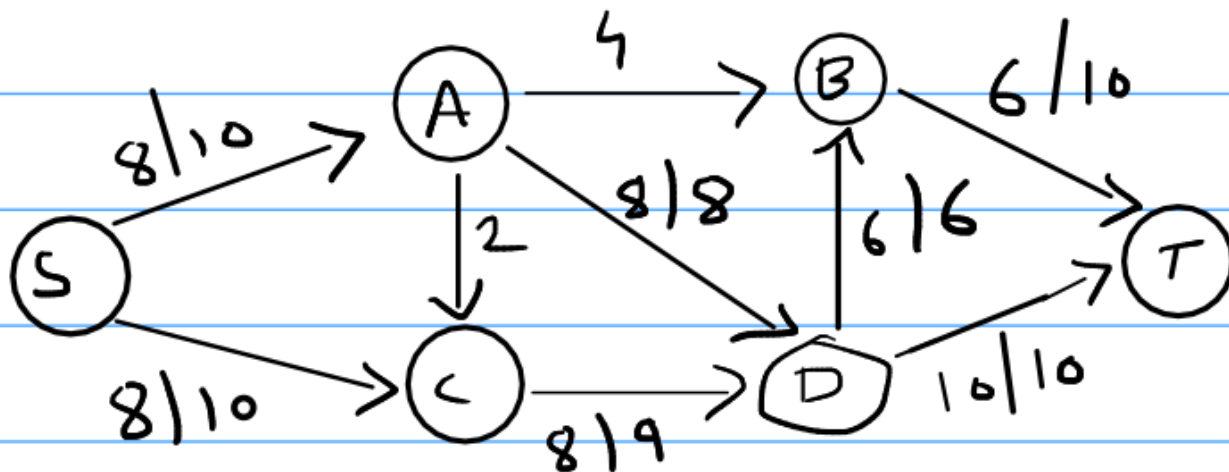


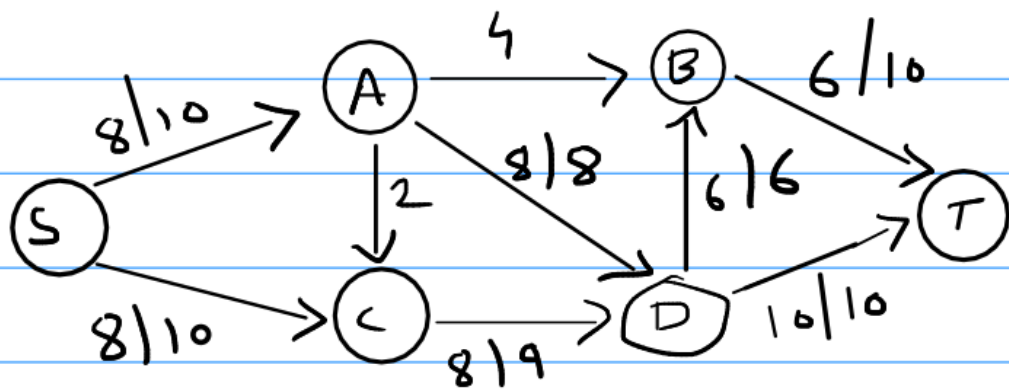
Augmenting Path	flow
S-A-D-T	8
S-C-D-T	2



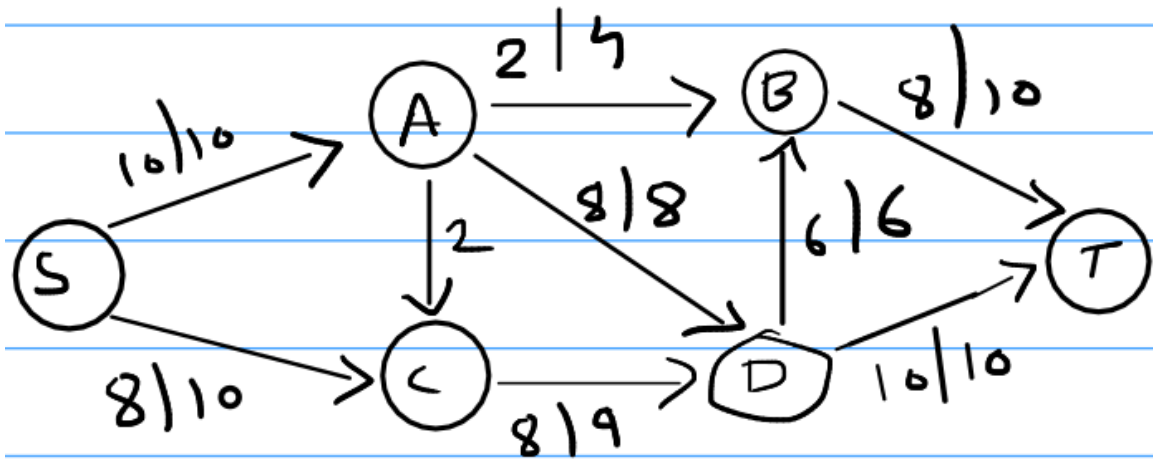


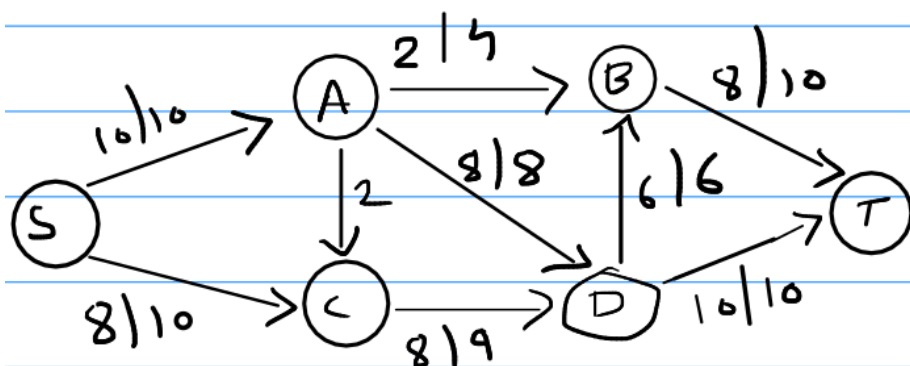
Augmenting Path	Flow
S-A-D-T	8
S-C-D-T	2
S-C-D-B-T	6



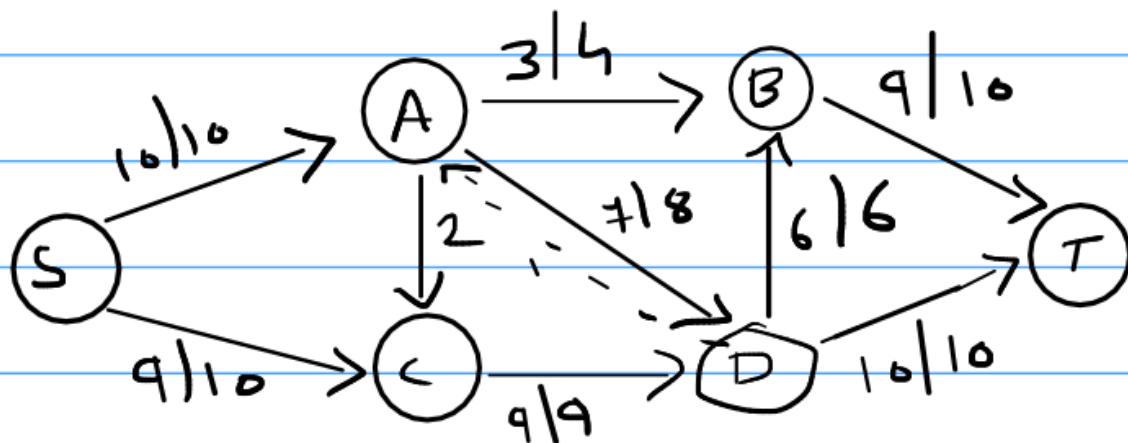


Augmenting Path	Flow
S-A-D-T	8
S-C-D-T	2
S-C-D-B-T	6
S-A-B-T	2



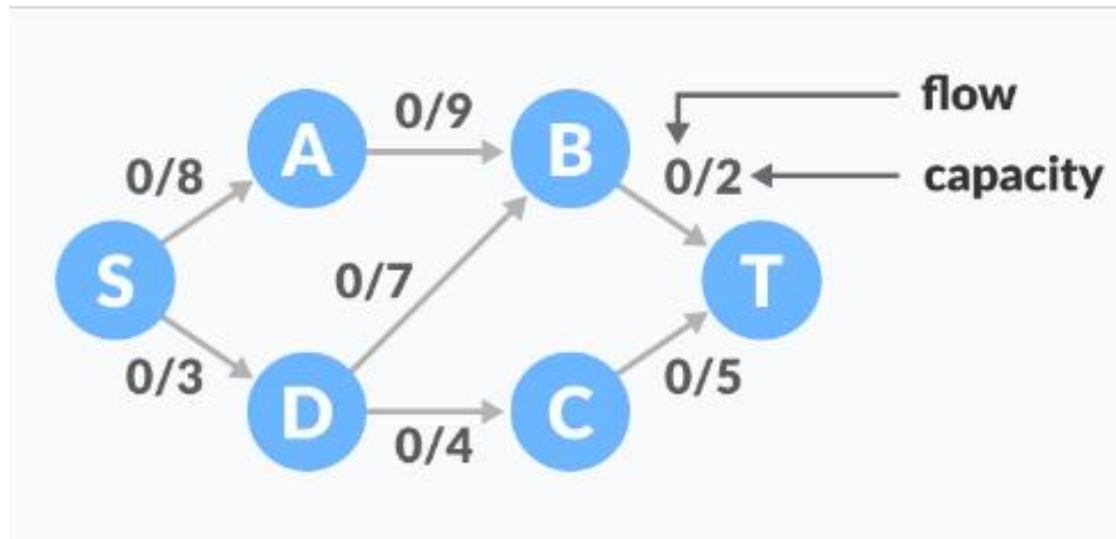


Augmenting Path	flow
S-A-D-T	8
S-C-D-T	2
S-C-D-B-T	6
S-A-B-T	2
S-C-D-A-B-T	1

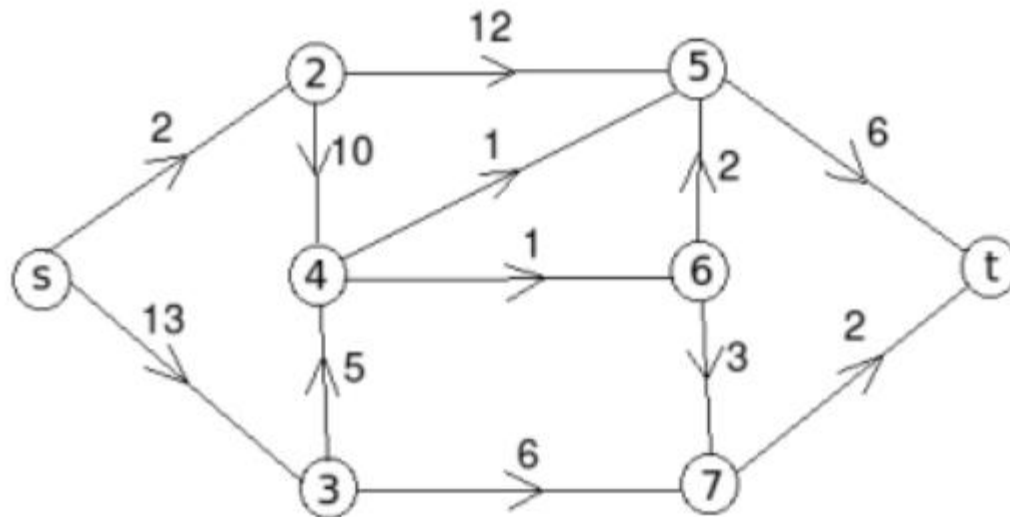


Augmenting Path	Flow
S-A-D-T	8
S-C-D-T	2
S-C-D-B-T	6
S-A-B-T	2
S-C-D-A-B-T	1
Total = 19	

Find the maximum flow Example-1



Find the maximum flow Example-2



Examples

- A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2, n vertices of degree 3. Determine the number of vertices and edges in tree.

By Handshaking theorem, Sum of the degree of Vertices = 2 Number of Edges

$$2n*1 + 3n*2 + n*3 = 2E$$

$$11n = 2E$$

$$E = 11n/2 \quad \dots(1)$$

As tree having, Number of Edges = Number of Vertices - 1

$$E = 2n + 3n + n - 1$$

$$E = 6n - 1 \quad \dots(2)$$

From 1 and 2

$$11n/2 = 6n - 1$$

$$11n = 12n - 2$$

$$n = 2$$

So Vertices = $6n = 12$

Edges = $6n - 1 = 11$

6.4 Let T be a tree with 50 edges. The removal of a certain edge from T yields two disjoint trees T_1 and T_2 . Given that the number of vertices in T_1 equals the number of edges in T_2 , determine the number of vertices and the number of edges in T_1 and T_2 .

$$v_1 + v_2 + v_3 + v_4 + \cdots + v_k = n_1 + n_2 = 51$$

$$5.4 \quad n_1 + n_2 = 51$$

$$n_1 = e_2 = n_2 - 1$$

Thus $n_1 = 25$, $e_1 = 24$, and $n_2 = 26$, $e_2 = 25$.

T_1 has 25 vertices 24 edges, while T_2 has 26 vertices and 25 edges.